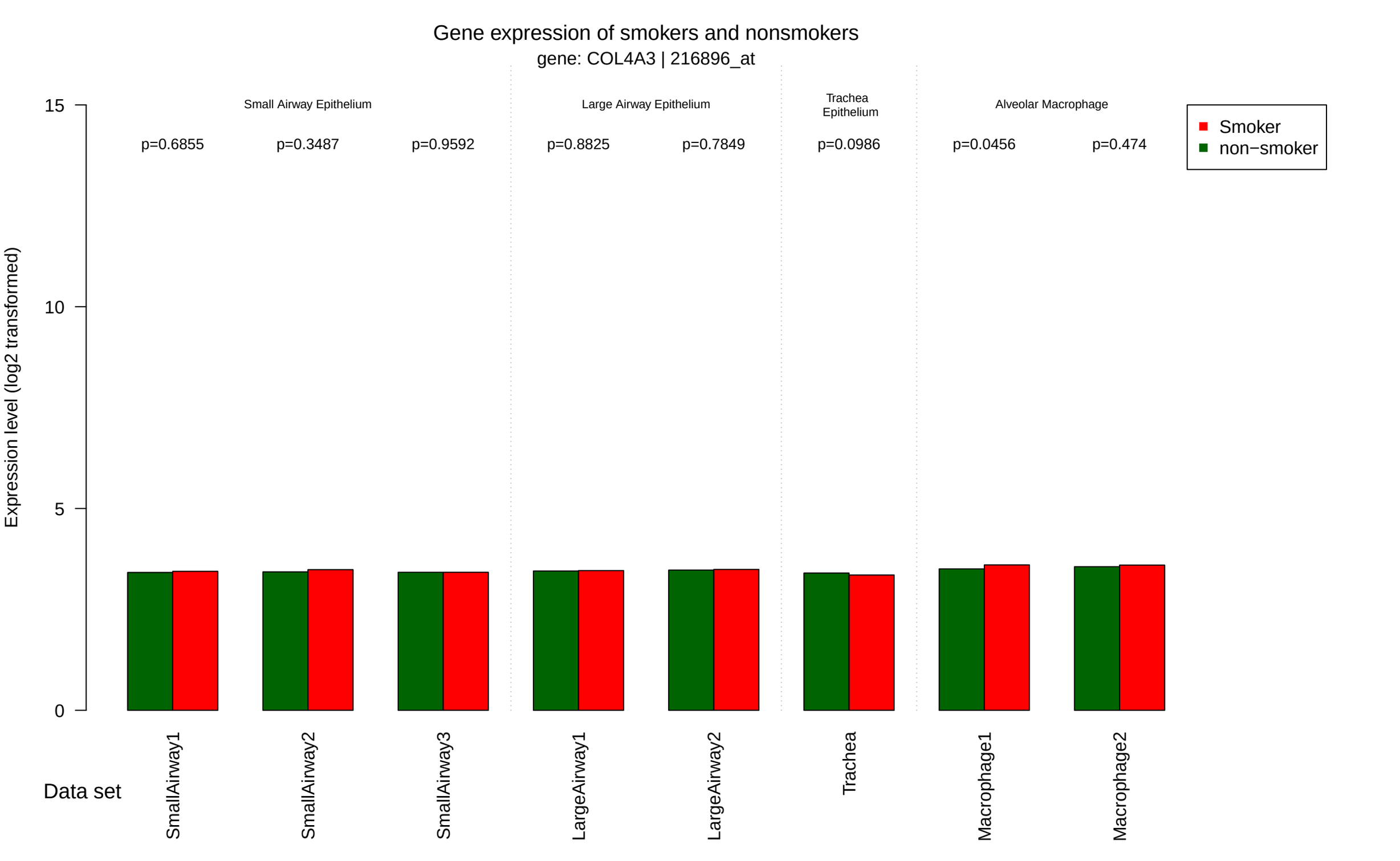
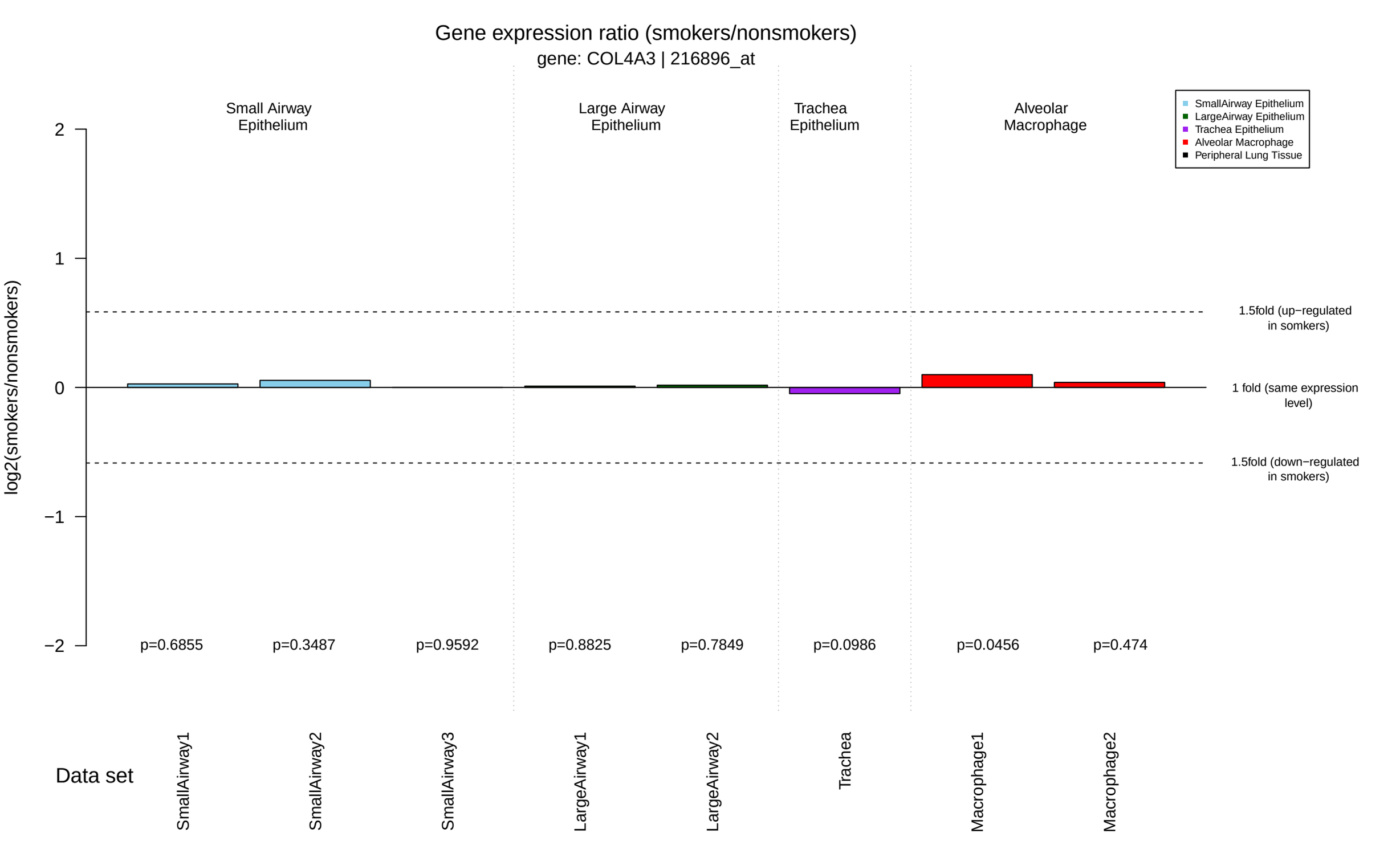
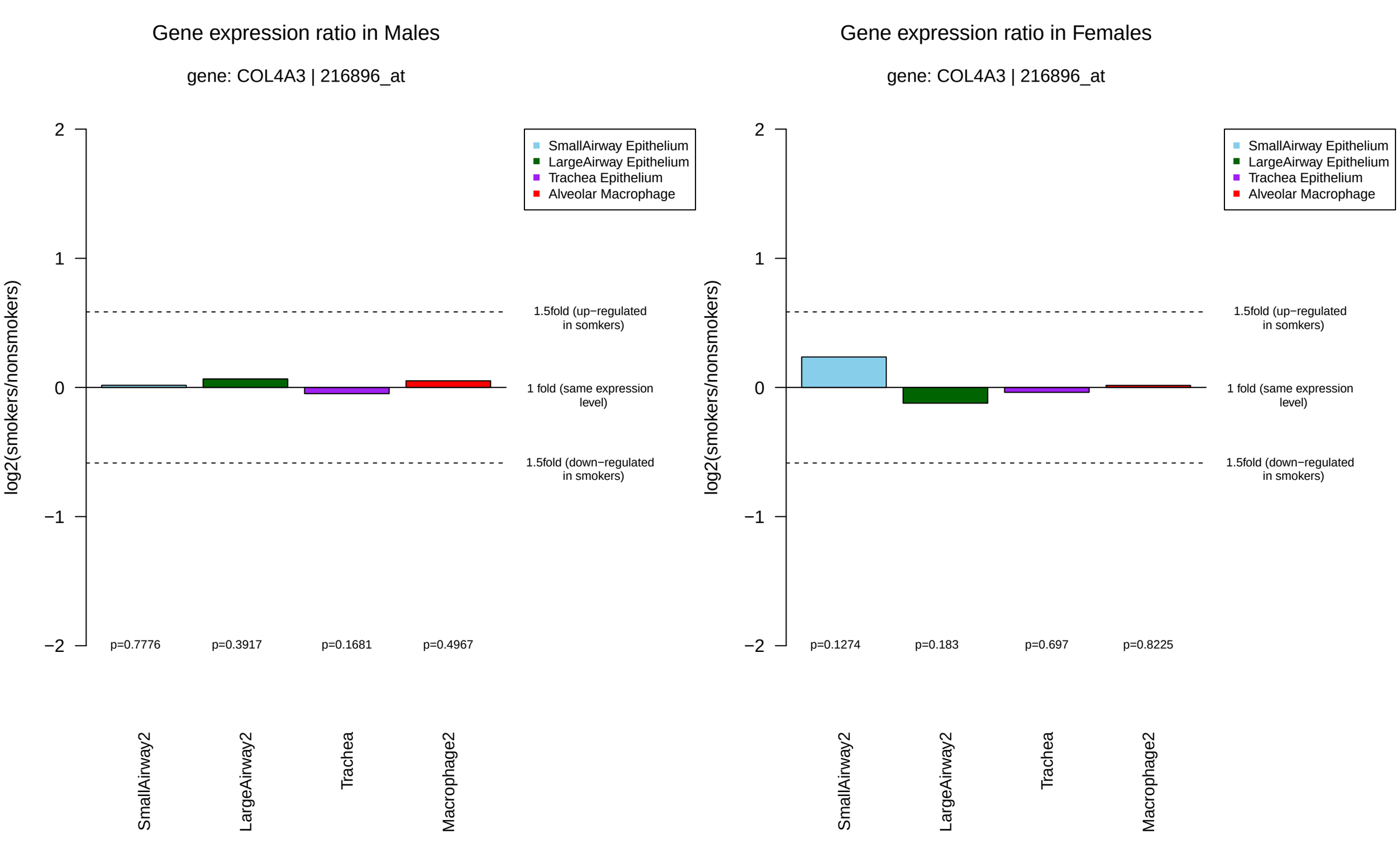
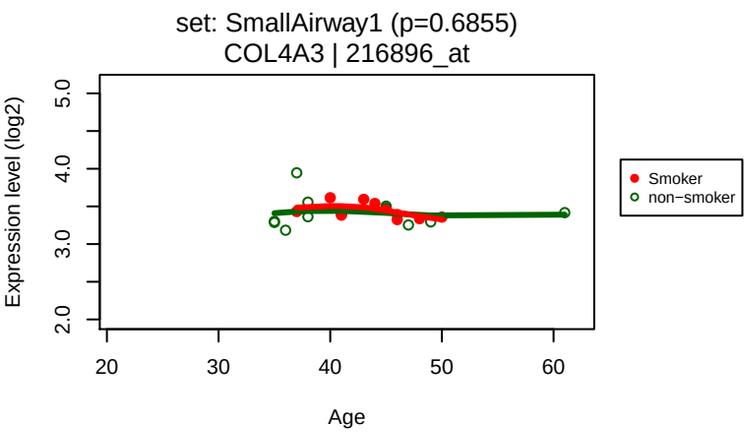
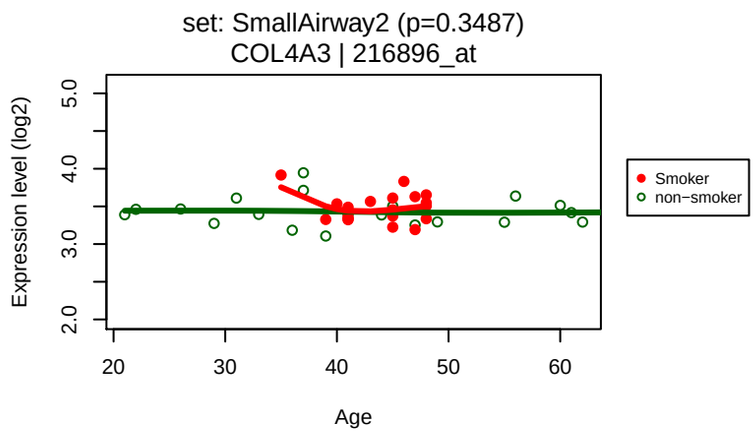
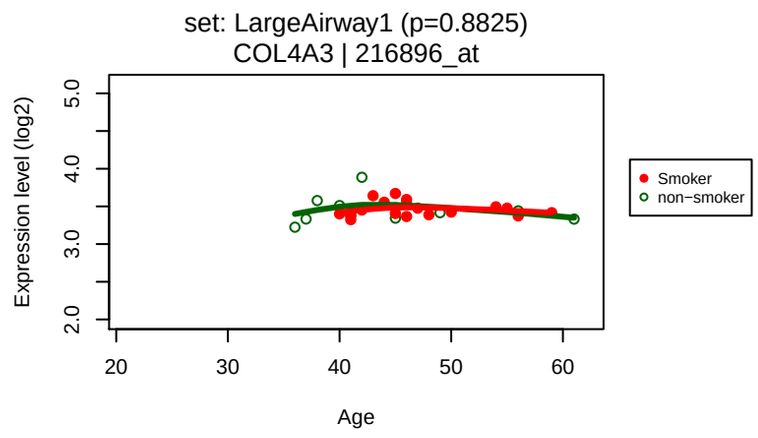

Gene expression of smokers and nonsmokers
gene: COL4A3 | 216896_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6855
p=0.3487
p=0.9592
p=0.8825
p=0.7849
p=0.0986
p=0.0456
p=0.474
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

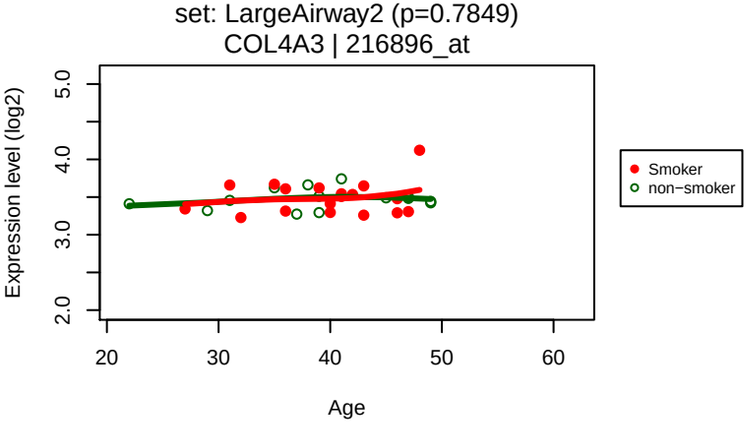

Gene expression ratio (smokers/nonsmokers)
gene: COL4A3 | 216896_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6855
p=0.3487
p=0.9592
p=0.8825
p=0.7849
p=0.0986
p=0.0456
p=0.474
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

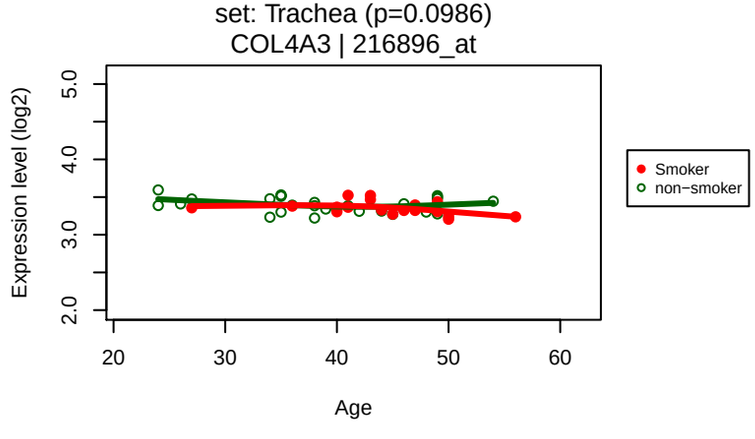

Gene expression ratio in Males
gene: COL4A3 | 216896_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7776
p=0.3917
p=0.1681
p=0.4967
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COL4A3 | 216896_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1274
p=0.183
p=0.697
p=0.8225
SmallAirway2
LargeAirway2
Trachea
Macrophage2

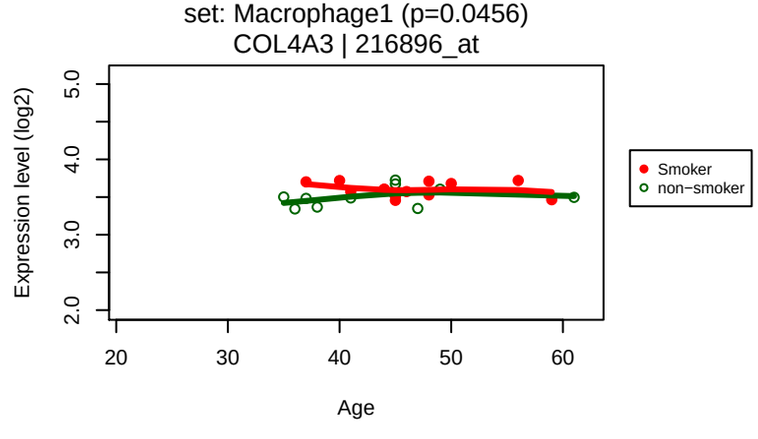

set: SmallAirway1 (p=0.6855)
COL4A3 | 216896_at
Expression level (log2)
Age
Smoker
non−smoker

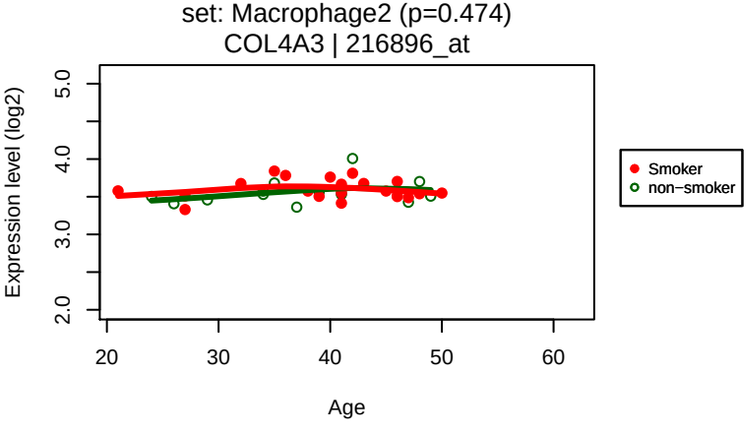

set: SmallAirway2 (p=0.3487)
COL4A3 | 216896_at
Expression level (log2)
Age
Smoker
non−smoker

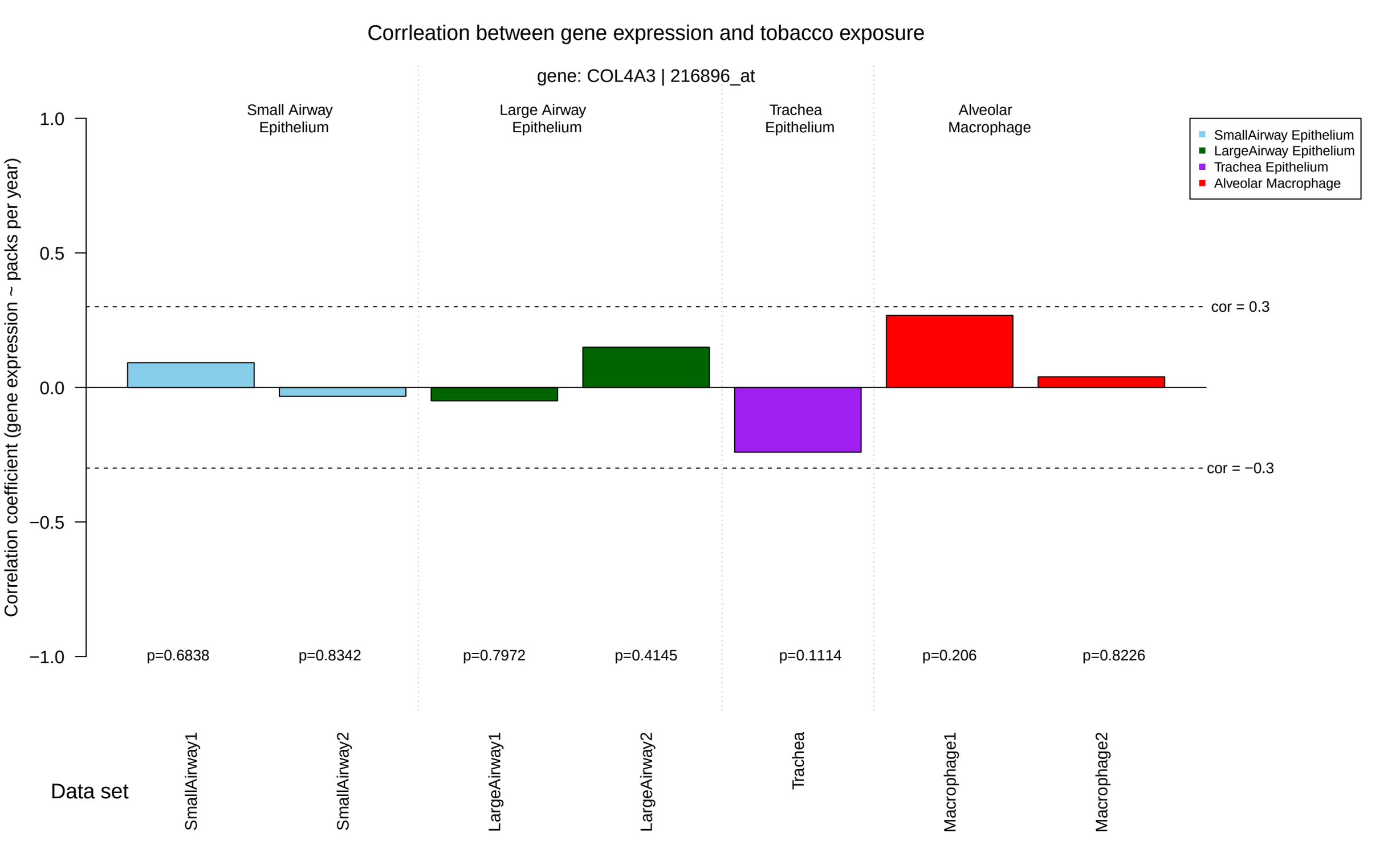

set: LargeAirway1 (p=0.8825)
COL4A3 | 216896_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7849)
COL4A3 | 216896_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0986)
COL4A3 | 216896_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0456)
COL4A3 | 216896_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.474)
COL4A3 | 216896_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COL4A3 | 216896_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6838
p=0.8342
p=0.7972
p=0.4145
p=0.1114
p=0.206
p=0.8226
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2